C. RS 422 Message Description

(1) During Automatic Mode of the Digital Cabin Pressure Control System (DCPCS), position data are exchanged as ASCII codes between the RS 422 interface of the controller and the respective electronic box of the outflow valve.

(2) For testing purposes, the RS 422 interface of the test equipment transmits opening and closing command signals to the active electronic box using command words as specified below.

(3) The RS 422 message is coded in ASCII, the codes represent characters or hex digits. It has the following sequence:

<message identifier> <data> <checksum>

(a) The first character of the message identifies the type of message, e.g.:

P - COMMANDED VALVE GATE POSITION
S - OFV STATUS
L - READ WORD REQUEST
K - READ BYTE REQUEST
M - WRITE BYTE REQUEST
etc.

(b) Messages with identifier Z are ignored (used in a special wraparound test).

(c) The length of the data section depends on the message type.

(d) The message is completed by transmission of the last two hex digits of checksum.

(4) The RS422 protocol of the electronic boxes (EBOX 1, EBOX 2) operates under the following transmission standards:

Mode:	8 Bit asynchronous (8N1)
Baud Rate:	9600 Baud
Updating:	20 Hz (every 20 ms)
Start Bit:	1
Stop Bit:	1
Parity:	None
Handshake:	None

(5) Examples of messages are given in Table 1002 and Table 1003 below.

(6) The L-message “READ WORD REQUEST” (Table 1002) is given as an example.

(7) The test is using P-messages and S-messages.

(a) The P-message “COMMANDED VALVE GATE POSITION” (Table 1003) is emitted from the controller and received by EBOX 1 or 2 to drive the valve gates to the commanded position.

(b) The S-message “OFV STATUS” is emitted from EBOX 1 or 2 and received by the controller. It contains the actual valve gate position as sensed by the feedback potentiometer.

CHARACTER NO.	CONTENTS	DESCRIPTION
1	Character L (Message Identifier)	READ WORD REQUEST (Read word value at specified address)
2	hex digit	bits 7...4 of address
3	hex digit	bits 3...0 of address
4	hex digit	bits 15...12 of address
5	hex digit	bits 11...8 of address
6	hex digit	bits 7...4 of checksum
7	hex digit	bits 3...0 of checksum

READ WORD REQUEST MESSAGE
TABLE 1002

CHARACTER NO.	CONTENTS	DESCRIPTION
1	Character P (Message Identifier)	COMMANDED VALVE GATE POSITION
2	hex digit	bits 7...4 of commanded position
3	hex digit	bits 3...0 of commanded position
4	hex digit	bits 15...12 of commanded position
5	hex digit	bits 11...8 of commanded position
6	hex digit	bits 7...4 of potentiometer calibration
7	hex digit	bits 3...0 of potentiometer calibration
8	hex digit	bits 15...12 of potentiometer calibration
9	hex digit	bits11...8 of potentiometer calibration
10	hex digit	bits 7...4 of checksum
11	hex digit	bits 3...0 of checksum

VALVE GATE POSITION COMMAND MESSAGE
TABLE 1003

CHARACTER NO.	CONTENTS	DESCRIPTION
1	Character S (Message Identifier)	OFV STATUS MESSAGE
2	hex digit	bits 7...4 of actuator state of operation
3	hex digit	bits 3...0 of actuator state of operation
4	hex digit	bits 7...4 of BIT FAULT MATRIX
5	hex digit	bits 3...0 of BIT FAULT MATRIX
6	hex digit	bits 15...12 of BIT FAULT MATRIX
7	hex digit	bits 11...8 of BIT FAULT MATRIX
8	hex digit	bits 7...4 of BIT FAULT MATRIX
9	hex digit	bits 3...0 of BIT FAULT MATRIX
10	hex digit	bits 15...12 of BIT FAULT MATRIX
11	hex digit	bits 11...8 of BIT FAULT MATRIX
12	hex digit	bits 7...4 of CURRENT OFV POSITION
13	hex digit	bits 3...0 of CURRENT OFV POSITION
14	hex digit	bits 15...12 of CURRENT OFV POSITION
15	hex digit	bits 11...8 of CURRENT OFV POSITION
16	hex digit	bits 7...4 of CHECKSUM
17	hex digit	bits 3...0 of CHECKSUM

OUTFLOW VALVE STATUS MESSAGE
TABLE 1004

BIT NO.	NAME	STATUS	
0	EBOX ON CHIP RAM FAIL	0	LSB
1	unused	0	
2	unused	0	D0
3	unused	0	
4	MOTOR DRIVE OVERCURRENT	0	
5	unused	0	
6	unused	0	D1
7	unused	0	
8	EBOX ON CHIP ROM FAIL	0	
9	RS422 WRAP AROUND FAIL	0	
10	RS422 CTR OFV ACTIVITY FAIL	0	D2
11	unused	0	
12	MOTOR DIRECTION WA FAIL	0	
13	SENSOR EXCITATION FAIL	0	
14	SENSOR RANGE FAIL	0	D3
15	unused	0	
16	unused	0	
17	LOOP CLOSURE FAIL	X	
18	unused	0	D4
19	unused	0	
20	unused	0	
21	unused	0	
22	unused	0	
23	unused	0	D5
24	UC AD CONVERTER FAIL	0	
25	MOTOR HALL SENSOR FAIL	0	
26	unused	0	
27	CAB PRESS SWITCH ACTIVATED	0	D6
28	RS422 MESSAGE FAIL	0	
29	unused	0	
30	unused	0	
31	unused	0	D7
	unused	0	MSB

NOTE: "X" signifies DON'T CARE Condition

ACTUATOR FAULT MATRIX
TABLE 1005

	CLOSING COMMAND		OPENING COMMAND	
CHARACTER NO.	ASCII HEX CODE	DECODED "FB00" (-6°)	ASCII HEX CODE	DECODED "7800" (120°)
1	50	P	50	P
2	30	0	30	0
3	30	0	30	0
4	46	F	37	7
5	42	B	38	8
6	30	0	30	0
7	30	0	30	0
8	30	0	30	0
9	30	0	30	0
10	46	F	44	D
11	38	8	46	F

VALVE GATE POSITION COMMAND MESSAGES USED IN TEST
TABLE 1006

D. Pin Assignment

(1) General

CONNECTOR	SPEC NUMBER	PLUG/SOCKET	CONNECTION	DESCRIPTION
J1	MS3114 E12-10P	plug	aircraft wiring	position feedback connector 1
J2	MS3114 E12-12P	plug	aircraft wiring	position feedback connector 2
J3	MS3112 E14-15S	socket	part of automotor	automotor connector
J4	MS3112 E16-26P	plug	aircraft wiring	EBOX Input from controller / feedback
J5	MS3122 E10-6P	plug	aircraft wiring	manual motor connector 1

ELECTRICAL CONNECTORS ON OUTFLOW VALVE
TABLE 1007

(4) Connector J3

PIN	WIRE NAME	SOURCE	DESTINATION	REMARKS
J3-A	PHASE1_OUT	EBOX	AMOT	
J3-B	PHASE2_OUT	EBOX	AMOT	
J3-C	PHASE3_OUT	EBOX	AMOT	
J3-D	HALL1_IN	AMOT	EBOX, Motor Tester	
J3-E	HALL2_IN	AMOT	EBOX, Motor Tester	
J3-F	HALL3_IN	AMOT	EBOX, Motor Tester	
J3-G	HALL_5VDC	EBOX, Motor Tester	AMOT	
J3-H	HALL_RTN	EBOX; Motor Tester	AMOT	
J3-J	not used			
J3-K	BRAKE_RTN	EBOX, Motor Tester	AMOT	
J3-L	not used			
J3-M	BRAKE_DRIVE_OUT	EBOX; Motor Tester	AMOT	
J3-N	not used			
J3-P	not used			
J3-R	not used			

PIN ASSIGNMENTS FOR EBOX CONNECTOR J3 (SOCKET)
TABLE 1010

(5) Connector J4

PIN	WIRE NAME	SOURCE	DESTINATION	REMARKS
J4-A	POT_EXC_ACT_OUT	EBOX	POT	Excitation voltage of feedback potentiometer for auto channel
J4-B	POT_RTN_ACT_OUT	EBOX	POT	Return voltage of feedback potentiometer for auto channel
J4-C	POT_WIP_ACT_IN	POT	EBOX	Wiper voltage of feedback potentiometer for auto channel
J4-D	not used			
J4-E	CHANNEL1_SEL	EBOX	FBK MOD	Input for Channel 1 select (former EBOX: RVT excitation)
J4-F	CHANNEL2_SEL	EBOX	FBK MOD	Input for Channel 2 select (former EBOX: RVT excitation)
J4-G	CHASSIS_GND	EBOX		chassis ground
J4-H	not used			
J4-J	PWR_28RTN	A/C	EBOX	
J4-K	PWR_28VDC	A/C	EBOX	
J4-L	DRIVE_ENABLE_PNL_ACT_IN	Panel	EBOX	
J4-M	RS422_ACT_CTR_B_OUT	EBOX	CTR	RS422 Output to controller
J4-N	RS422_ACT_CTR_A_OUT	EBOX	CTR	RS422 Output to controller
J4-P	RS422_CTR_ACT_B_IN	CTR	EBOX	RS422 Input in EBOX
J4-R	RS422_CTR_ACT_A_IN	CTR	EBOX	RS422 Input in EBOX
J4-S	not used			
J4-T	not used			
J4-U	not used			
J4-V	not used			
J4-W	SIGNAL_GND	EBOX		Ground
J4-X				not used
J4-Y				not used

PIN ASSIGNMENT FOR EBOX CONNECTOR J4 (SOCKET)
TABLE 1011 (SHEET 1 OF 2)

(3) Connector J2

PIN	WIRE NAME	SOURCE	DESTINATION	REMARKS
J2-A	POT_EXC_ACT2_IN	EBOX2	POT2	Excitation voltage of feedback potentiometer for auto channel 2
J2-B	POT_RTN_ACT2_IN	EBOX2	POT2	Return voltage of feedback potentiometer for auto channel 2
J2-C	POT_WIP_ACT2_OUT	POT2	EBOX2	Wiper voltage of feedback potentiometer for auto channel 2
J2-D	not used			
J2-E	CHANNEL1_SEL		EBOX2	Input for Channel 1 select, opened; (in former EBOX RVT Excitation High)
J2-F	CHANNEL2_SEL		EBOX2	Input for Channel 2 select, connected to J2.B; (in former EBOX RVT Excitation Low)
J2-G	not used			
J2-H	CHASSIS_GND			Chassis Ground
J2-J	not used			
J2-K	not used			
J2-L	not used			
J2-M	not used			

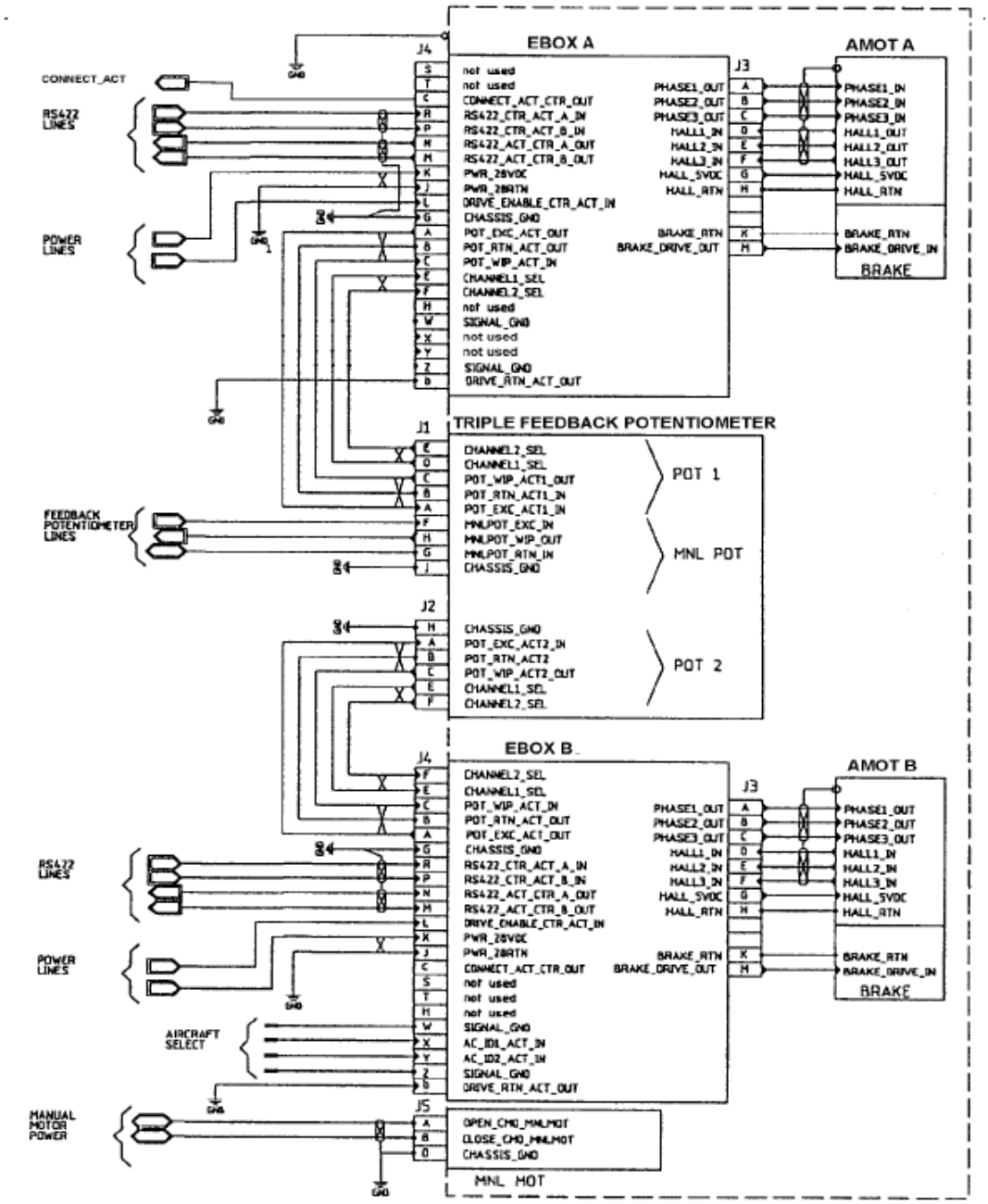
PIN ASSIGNMENTS FOR TRIPLE POTENTIOMETER CONNECTOR J2 (SOCKET)
TABLE 1009

(2) Connector J1

PIN	WIRE NAME	SOURCE	DESTINATION	REMARKS
J1-A	POT_EXC_ACT1_IN	EBOX1	POT1	Excitation voltage of feedback potentiometer for auto channel 1
J1-B	POT_RTN_ACT1_IN	EBOX1	POT1	Return voltage of feedback potentiometer for auto channel 1
J1-C	POT_WIP_ACT1_OUT	POT1	EBOX1	Wiper voltage of feedback potentiometer for auto channel 1
J1-D	CHANNEL1_SEL		EBOX1	Input for Channel 1 select, connected to J1.B; (in former EBOX RVT Excitation High)
J1-E	CHANNEL2_SEL		EBOX1	Input for Channel 2 select, opened; (in former EBOX RVT Excitation Low)
J1-F	MNLPOT_EXC_IN	Panel	MNLPOT	Excitation voltage of feedback potentiometer for manual backup channel
J1-G	MNLPOT_RTN_IN	Panel	MNLPOT	Return voltage of feedback potentiometer for manual backup channel
J1-H	MNLPOT_WIP_OUT	MNLPOT	Panel	Wiper voltage of feedback potentiometer for manual backup channel
J1-J	CHASSIS GND			Chassis ground
J1-K	not used			

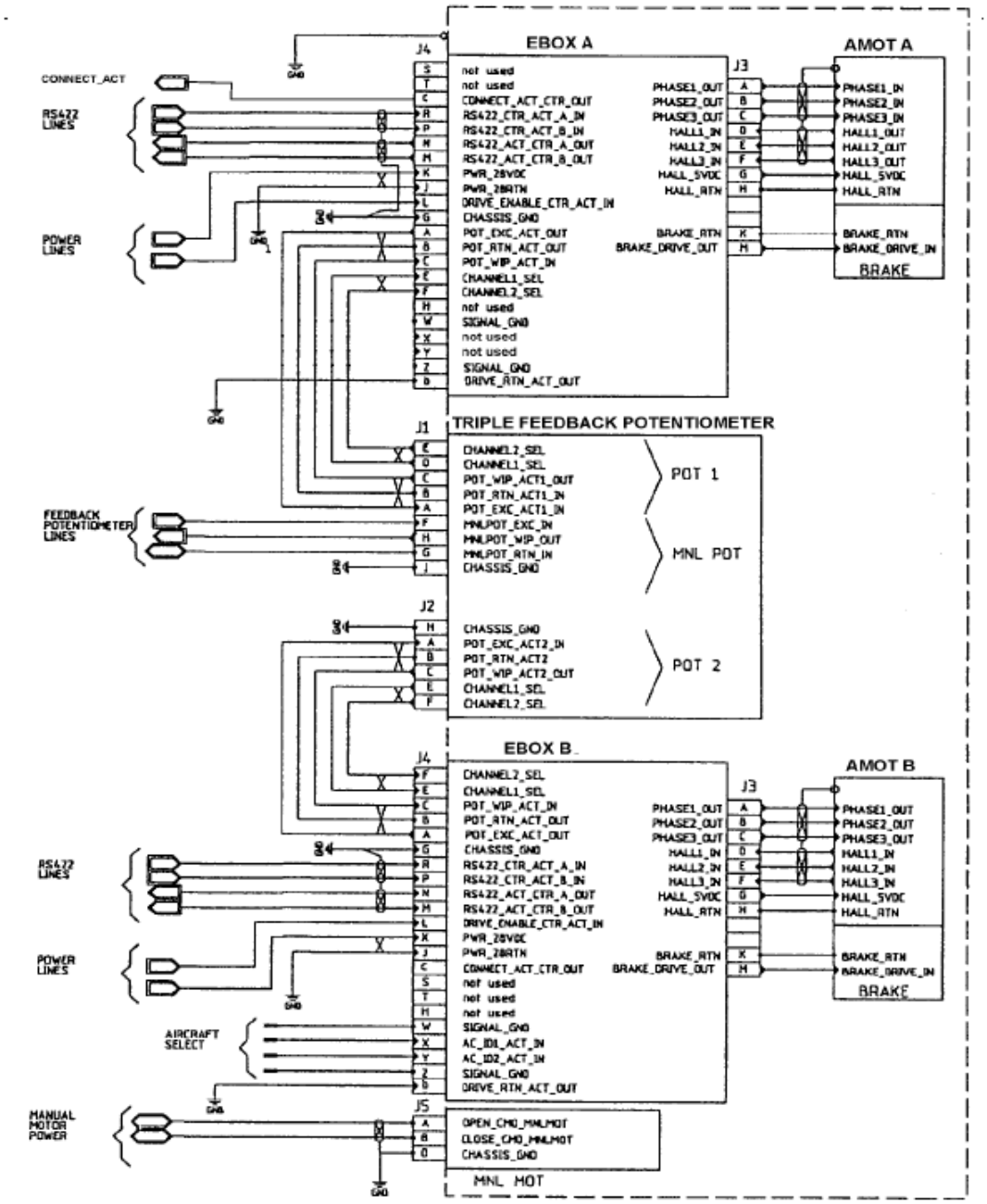
PIN ASSIGNMENTS FOR TRIPLE POTENTIOMETER CONNECTOR J1 (SOCKET)
TABLE 1008

E. Test Setup



INTERCONNECTION DIAGRAM
FIGURE 1001

E. Test Setup



INTERCONNECTION DIAGRAM
FIGURE 1001